



科学写作与报告

期刊分区与论文分类
主讲人：白琳

本课程的基本内容:

- 期刊分区
- 论文分类——综述论文
- 论文分类——科技论文
- 论文分类——学位论文

期刊

■ 什么是期刊？

- 期刊，定期出版的刊物。如周刊、旬刊、半月刊、月刊、季刊、半年刊、年刊等。
- 按照内容可以将期刊分为一般期刊、学术期刊、行业期刊、检索期刊。
- 本课程中所讲的期刊特指**学术期刊**。

论文的定义

■ 国家标准（GB 7718-87）对于学术论文的定义

“学术论文是某一学术课题在实验性、理论性或观测性上具有新的科学研究成果或创新见解和知识的科学记录；或是某种已知原理应用于实际中取得新进展的科学总结，用以提供学术会议上宣读、交流或讨论；或在学术刊物上发表；或作其他用途的书面文件”

中华人民共和国国家标准

科学技术报告、学位论文和
学术论文的编写格式

Presentation of scientific and
technical reports, dissertations
and scientific papers

UDC 001.81

GB 7718—87

2.3 学术论文

学术论文是某一学术课题在实验性、理论性或观测性上具有新的科学研究成果或创新见解和知识的科学记录；或是某种已知原理应用于实际中取得新进展的科学总结，用以提供学术会议上宣读、交流或讨论；或在学术刊物上发表；或作其他用途的书面文件。

学术论文应提供新的科技信息，其内容应有所发现、有所发明、有所创造、有所前进，而不是重

国家标准局1987-05-05批准

1988-01-01实施

期刊分区

■ 为什么要进行期刊分区？

- 目前SCI数据库收录了9940种期刊，SCI库的期刊质量和影响力存在很大的差距，尤其是不同学科之间的SCI期刊，很难进行比较和评价。根据学科内部杂志的真实水平对杂志分档次，可以避免学科横向比较时受影响因子的不客观影响，兼顾学科公平，更科学地对学术期刊进行评价。

期刊分区

■ 国内主流参考的**SCI**分区

- 中科院JCR分区
- Journal Ranking分区

■ 中科院**JCR**分区

- 是中国科学院国家科学图书馆世界科学前沿分析中心（原中国科学院文献情报中心）的研究成果，对目前**SCI**核心库加上扩展库期刊的影响力等因素，以年度和学科为单位，对**SCI**期刊进行4个等级的划分。

期刊分区

■ 中科院JCR分区方法

- 基于一个期刊的学术影响力（影响因子，impact factor，IF），主要采用的三年平均IF，各**大类**学科排序，前5%为该类1区、6%～20%为2区、21%～50%为3区，其余为4区。

■ 科睿唯安JCR的Journal Ranking分区

- 没有设置大类学科，只分为176个中科院分区小类学科。
- 以当年的IF为基础，均匀分区(每一区占比25%)。

■ 影响因子

- 代表期刊影响大小的一项定量指标，也就是某刊平均每篇论文的被引用数。
- 计算方法：
 - 某刊在某年被全部源刊物**引证**该刊前两年发表论文的**次数**/该刊前两年所发表的全部源**论文数**

期刊分区

■ 中科院JCR分区查询路径

❑ <http://www.fenqubiao.com>

❑ 图书馆资源直达

首页 > 数据库导航

中国科学院文献情报中心期刊分区表
CAS Journal Ranking

中科院期刊分区表
别名：中国科学院文献情报中心期刊分区表, CAS Journal Ranking, JCR, fq, zky

资源打分 评论咨询 故障申报

根据IP推荐的访问地址 学院路校区 10.193.106.43 【校园网IP, 可直接访问所有电子资源】

↪ 校内直达

基本信息

资源类型：文摘, 期刊 采购方式：正式开通 语言：中文 学科分类：综合, 航空航天, 材料/化学/环境/生医, 电子/机械/计算机/控制/交通, 数学/物理, 人文社科, 其它
咨询方式：孙老师, 010-82339288

数据库简介

采购范围：2019年-2023年/2025年, 账号密码：BUAAFQB/xy137buaa

中国科学院文献情报中心期刊分区表（CAS Journal Ranking, 以下简称“期刊分区表”）是中国科学院文献情报中心计量与评价部的研究成果。“期刊分

期刊分区

- 科睿唯安JCR的Journal Ranking分区查询路径
 - 登录Web of Science，直接搜索期刊名

部分通信类期刊Journal Ranking分区及影响因子

期刊名称	分区（影响因子）
IEEE Wireless Communications	Q1 (IF=11.5)
IEEE Network	Q1 (IF=6.3)
IEEE Communications Magazine	Q1 (IF=8.2)
IEEE Journal on Selected Areas in Communications	Q1 (IF=17.2)
IEEE Transactions on Wireless Communications	Q1 (IF=10.7)
IEEE Transactions on Vehicular Technology	Q1 (IF=7.1)
IEEE Transactions on Communications	Q1 (IF=8.3)
IEEE Transactions on Signal Processing	Q1 (IF=5.8)
IEEE Transactions on Antennas and Propagation	Q1 (IF=5.8)
Sensors	Q2 (IF=3.5)
IEEE Communications Letters	Q2 (IF=4.4)
IEEE Access	Q2 (IF=3.6)
EURASIP Journal on Wireless Communications and Networking	Q3 (IF=1.9)

期刊分区

- IEEE Wireless Communications
- ISSN号、EISSN号、缩写、所属学科类别、论文语言、期刊属国、期刊频次等

Home > Journal profile

JCR Year
2024

Favorite Export

IEEE WIRELESS COMMUNICATIONS

View title change

ISSN

1536-1284

EISSN

1558-0687

JCR ABBREVIATION

IEEE WIREL COMMUN

ISO ABBREVIATION

IEEE Wirel. Commun.

期刊名缩写

Journal information

EDITION

Science Citation Index Expanded (SCIE)

CATEGORY

TELECOMMUNICATIONS

COMPUTER SCIENCE, HARDWARE & ARCHITECTURE

COMPUTER SCIENCE, INFORMATION SYSTEMS

ENGINEERING, ELECTRICAL & ELECTRONIC

所属学科

LANGUAGES

English

语言

REGION

USA

1ST ELECTRONIC JCR YEAR

2002

Publisher information

PUBLISHER

IEEE-INST
ELECTRICAL
ELECTRONICS
ENGINEERS INC

ADDRESS

445 HOES LANE,
PISCATAWAY, NJ
08855-4141

PUBLICATION FREQUENCY

6 issues/year

频次

期刊分区

- IEEE Wireless Communications
- Journal Impact Factor(JIF, 期刊影响因子)
- 2024年影响因子、近5年影响因子、影响因子贡献论文名单

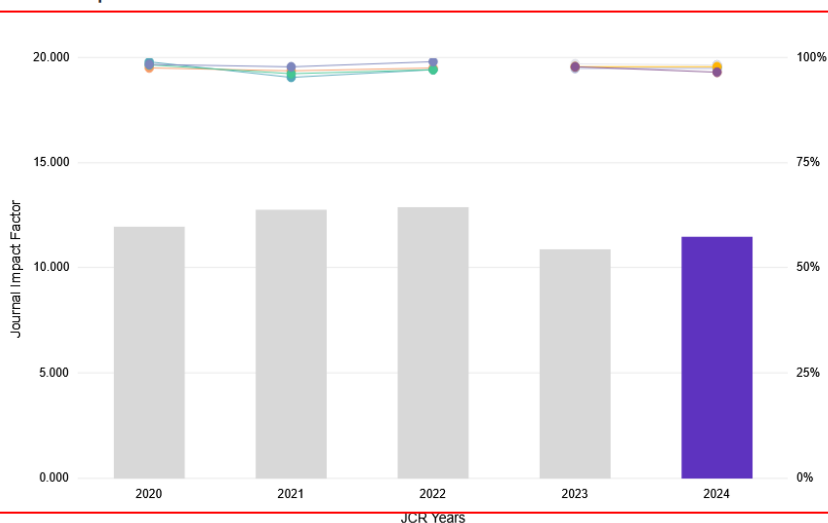
Journal Impact Factor

The Journal Impact Factor (JIF) is a journal-level metric calculated from data indexed in the Web of Science Core Collection. It should be used with careful attention to the many factors that influence citation rates, such as the volume of publication and citations characteristics of the subject area and type of journal. The Journal Impact Factor can complement expert opinion and informed peer review. In the case of academic evaluation for tenure, it is inappropriate to use a journal-level metric as a proxy measure for individual researchers, institutions, or articles. [Learn more](#)

2024 JOURNAL IMPACT FACTOR
11.5
[View calculation](#)

JOURNAL IMPACT FACTOR WITHOUT SELF CITATIONS
11.1
[View calculation](#)

2024影响因子
Journal Impact Factor Trend 2024



近五年影响因子及趋势

Journal Impact Factor contributing items

[Export](#)

Citable items (241)		Citing Sources (440)
TITLE	CITATION COUNT	
Semantic Communications: Overview, Open Issues, and Future Research Directions	147	
AI-Native Network Slicing for 6G Networks	81	
Integrated Sensing and Communication with Reconfigurable Intelligent Surfaces: Opportunities, Applications, and Future Directions	66	
The Roadmap of Communication and Networking in 6G for the Metaverse	65	
DEVELOPING NOMA TO NEXT GENERATION MULTIPLE ACCESS: FUTURE VISION AND RESEARCH OPPORTUNITIES	57	
Satellite-Based Computing Networks with Federated Learning	46	
Task-Oriented Communications for 6G: Vision, Principles, and Technologies	44	
Decentralized P2P Federated Learning for Privacy-Preserving and Resilient Mobile Robotic Systems	42	
Reconfigurable Intelligent Surface-Aided Multi-User Networks: Interplay Between NOMA and RIS	40	

贡献论文名单及被引次数

期刊分区

- ❑ **Journal Citation Indicator(JCI, 期刊引文指标)**是科睿唯安公司于**2021**年推出的期刊评价指标，用于衡量期刊近期出版物的引文影响力。
- ❑ **JCI**指期刊在前三年期间发表的文章和综述的学科规范化引文影响力的平均值。**1.0**代表的是全球平均水平。如一本期刊的**JCI**为**1.3**, 则说明它的引用影响力比该类别的平均水平高 **30%**。
- ❑ 引入了学科规范化引文影响力（**CNCI**），对学科类别，发表年份，文章类型进行了标准化。
- ❑ 不仅可以反映期刊引文影响力与平均引文影响力的关系，而且实现了期刊层面的跨学科比较。

期刊分区

- ❑ IEEE Wireless Communications
- ❑ JCI值2.57，总被引次数12680

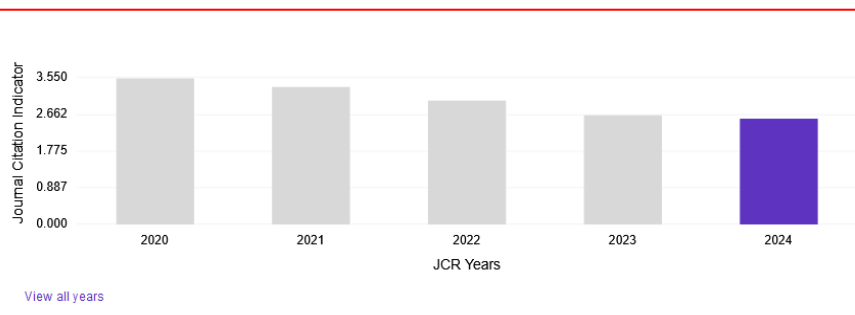
Journal Citation Indicator (JCI)

2.57

JCI值

Export

The Journal Citation Indicator (JCI) is the average Category Normalized Citation Impact (CNCI) of citable items (articles & reviews) published by a journal over a recent three year period. The average JCI in a category is 1. Journals with a JCI of 1.5 have 50% more citation impact than the average in that category. It may be used alongside other metrics to help you evaluate journals. [Learn more](#)



JCI近五年趋势

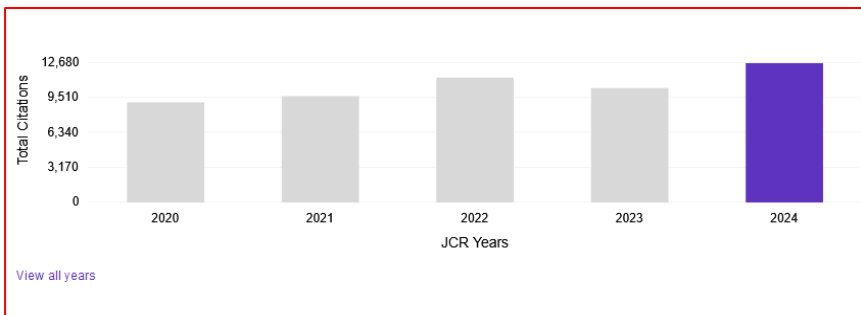
Total Citations

12,680

总引次数

Export

The total number of times that a journal has been cited by all journals included in the database in the JCR year. Citations to journals listed in JCR are compiled annually from the JCR years combined database, regardless of which JCR edition lists the journal.

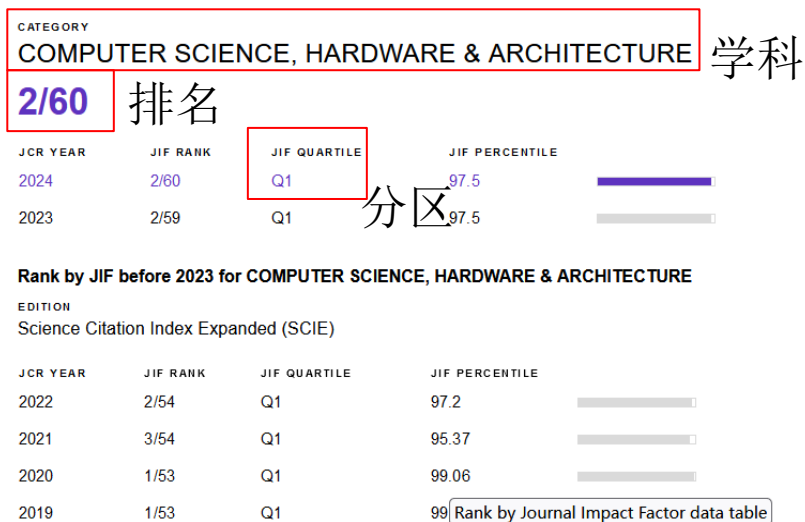


近五年总被引趋势

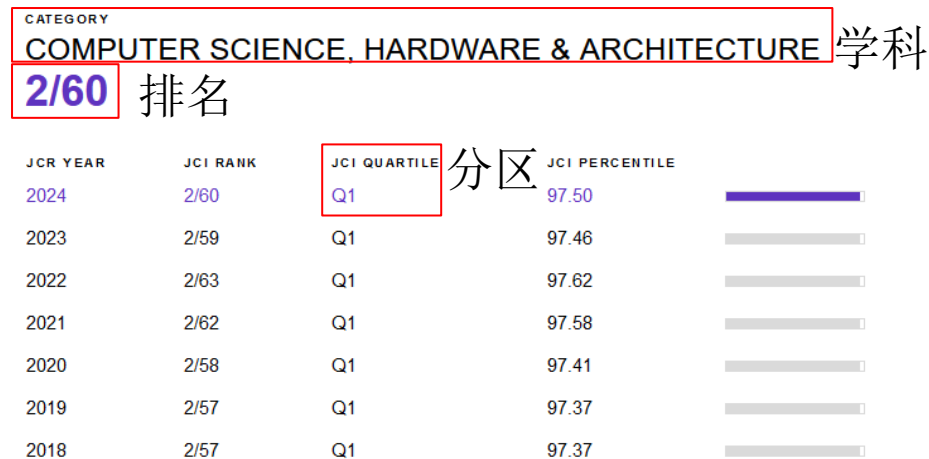
期刊分区

IEEE Wireless Communications

按影响因子排名



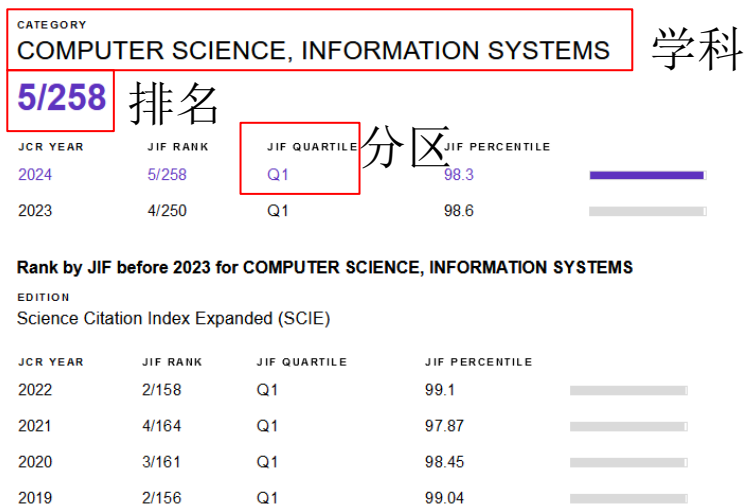
按期刊引文指标排名



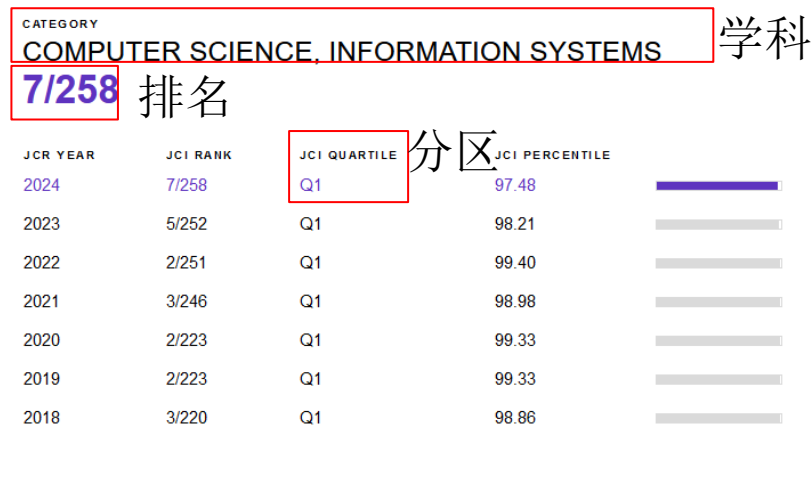
期刊分区

IEEE Wireless Communications

按影响因子排名



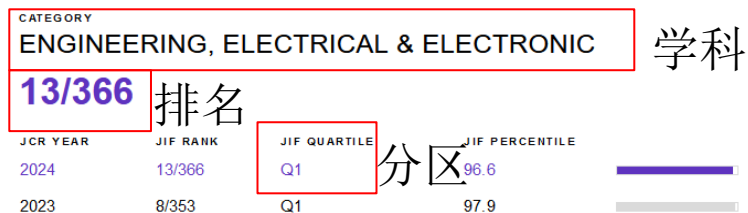
按期刊引文指标排名



期刊分区

IEEE Wireless Communications

按影响因子排名



Rank by JIF before 2023 for ENGINEERING, ELECTRICAL & ELECTRONIC

EDITION

Science Citation Index Expanded (SCIE)

JCR YEAR	JIF RANK	JIF QUARTILE	JIF PERCENTILE
2022	7/275	Q1	97.6
2021	9/276	Q1	96.92
2020	7/273	Q1	97.62
2019	5/266	Q1	98.31

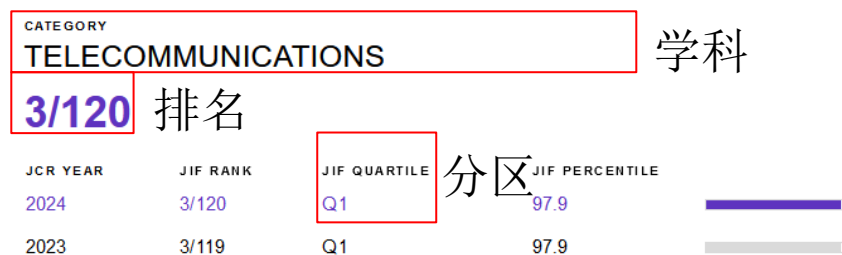
按期刊引文指标排名



期刊分区

IEEE Wireless Communications

按影响因子排名



Rank by JIF before 2023 for TELECOMMUNICATIONS

EDITION
Science Citation Index Expanded (SCIE)

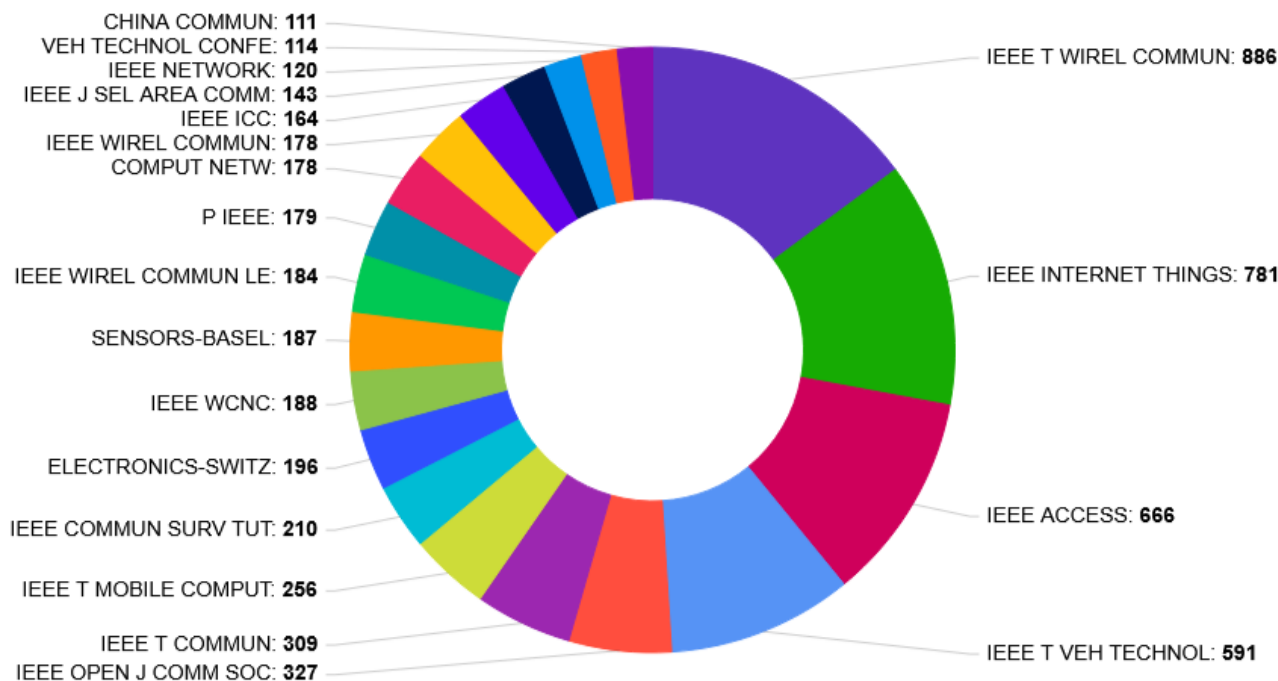
JCR YEAR	JIF RANK	JIF QUARTILE	JIF PERCENTILE
2022	3/88	Q1	97.2
2021	4/93	Q1	96.24
2020	2/91	Q1	98.35
2019	3/90	Q1	97.22

按期刊引文指标排名



期刊分区

- IEEE Wireless Communications
- 期刊引用关系：引用IEEE Wireless Communications的期刊前20名



期刊分区

■ 国际期刊预警名单

- 近十年间全球学术不端引发的撤稿中，中国占比从**14.4%**（2011年）增加至**53.2%**（2020年）。
- 2018年，中共中央办公厅、国务院办公厅印发《关于进一步加强科研诚信建设的若干意见》，明确指出“支持相关机构发布国内和国际学术期刊预警名单....”。
- 综合评判期刊载文量、作者国际化程度、拒稿率、论文处理费（APC）、期刊超越指数、自引率、撤稿信息等，找出那些具备风险特征、具有潜在质量问题的学术期刊。
- 依据各刊数据差异，将预警级别分为**高、中、低**三档，风险指数依次减弱。
- 《国际期刊预警名单》旨在提醒科研人员审慎选择成果发表平台、提示出版机构强化期刊质量管理。

期刊分区

- 国际期刊预警名单
- 查询方式: <https://ewl.fenqubiao.com/#/>

2025年《国际期刊预警名单》

2025年3月19日中国科学院文献情报中心-期刊分区表团队发布

期刊	ISSN/ EISSN	预警原因
Wireless Personal Communications	0929-6212	论文工厂
Natural Resources Forum	0165-0203	论文工厂
Computers & Electrical Engineering	0045-7906	论文工厂
NUMERICAL HEAT TRANSFER PART A-APPLICATIONS	1040-7782	论文工厂
SCALABLE COMPUTING-PRACTICE AND EXPERIENCE	1895-1767	论文工厂

期刊分区

■ 2024年部分国际期刊预警名单

2024年2月19日中国科学院文献情报中心-期刊分区表团队发布

期刊	ISSN/ EISSN	预警原因
CANCERS	2072-6694	引用操纵
DIAGNOSTICS	2075-4418	引用操纵
ENVIRONMENTAL SCIENCE AND POLLUTION RESEARCH	0944-1344	引用操纵、论文工厂
FUEL	0016-2361	引用操纵
JOURNAL OF CLINICAL MEDICINE	2077-0383	引用操纵
JOURNAL OF PERSONALIZED MEDICINE	2075-4426	引用操纵
RADIOLOGIA MEDICA	0033-8362	引用操纵
BIOENGINEERED	2165-5979	论文工厂
CONNECTION SCIENCE	0954-0091	论文工厂
MULTIMEDIA TOOLS AND APPLICATIONS	1380-7501	论文工厂
PSYCHIATRIA DANUBINA	0353-5053	论文工厂
JOURNAL OF BIOBASED MATERIALS AND BIOENERGY	1556-6560	论文工厂、中国作者占比畸高
JOURNAL OF BIOMATERIALS AND TISSUE ENGINEERING	2157-9083	论文工厂、中国作者占比畸高
JOURNAL OF BIOMEDICAL NANOTECHNOLOGY	1550-7033	论文工厂、中国作者占比畸高
JOURNAL OF NANOELECTRONICS AND OPTOELECTRONICS	1555-130X	论文工厂、中国作者占比畸高
JOURNAL OF SENSORS	1687-725X	论文工厂、中国作者占比畸高
MATERIALS EXPRESS	2158-5849	论文工厂、中国作者占比畸高
SCIENCE OF ADVANCED MATERIALS	1947-2935	论文工厂、中国作者占比畸高
ALTERNATIVE THERAPIES IN HEALTH AND MEDICINE	1078-6791	中国作者占比畸高
CMES-COMPUTER MODELING IN ENGINEERING & SCIENCES	1526-1492	中国作者占比畸高
EXPERIMENTAL AND THERAPEUTIC MEDICINE	1792-0981	中国作者占比畸高
FRONTIERS IN ENERGY RESEARCH	2296-598X	中国作者占比畸高

论文分类

- 综述报告，研究型科技论文，学位论文
- 综述报告
 - 综述报告，或者说文献综述，是科学研究的调研阶段不可缺少的部分。与研究型论文不同，撰写综述报告的目的在于充分调研了解（或使其他研究者得以了解）待研究领域的基础背景知识和最新研究成果，以保证未来研究课题的独创性以及可实现性。

论文分类——综述报告

■ 综述论文（Survey paper）

- 很多期刊都包含一定数量的综述类论文。与通常意义上的综述报告不同，综述类论文着眼点或是更加深入聚焦，或是更加系统全面。通常包括研究背景、研究目标、研究动态，关键成果和评价，以及对未来方向的展望等方面。
- 举例：
 - IEEE Communication Magazine
 - IEEE Communications Surveys & Tutorials

论文分类——综述报告

■ IEEE Communication Magazine

■ 期刊要求

聚焦于最新的研究点，
基于以往的研究成果，
提出和阐述研究领域可能的最新技术和创新点。
（论文篇幅短小精炼）

IEEE Communications Magazine

This award-winning magazine brings you the latest international coverage of current issues and advances in key areas of wireless, optical and wired communications. Written in tutorial applications-driven style by the industry's leading experts, **IEEE Communications Magazine** delivers practical, current information on hot topics, implementations, and best industry practices.

Considered by many to be their most important member benefit, monthly articles feature in-depth descriptions of technologies, systems, services, market trends, development methods, regulatory and policy issues, and significant global events. These articles are complemented by a variety of departments, including: The President's Page, Conference Calendar, Book Reviews, the Global Communications Newsletter, Society News, History of Communications, and more. This monthly magazine becomes your single most important source of state-of-the-science communications information.

To become a volunteer of the **IEEE Communications Magazine** Editorial Board, please fill out the [online application](#).

Considered by many to be their most important member benefit, monthly articles feature in-depth descriptions of technologies, systems, services, market trends, development methods, regulatory and policy issues, and significant global events. These articles are complemented by a variety of

论文分类——综述报告

IEEE Communications Surveys & Tutorials

■ 期刊要求

提供该研究领域全面而广泛介绍与总结，包括研究发展，过去与现在的代表性成果总结，以及对未来研究趋势的展望和分析。

（论文篇幅较长）

IEEE Communications Surveys & Tutorials

The *IEEE Communications Surveys & Tutorials* is an online journal published by the IEEE Communications Society for tutorials and surveys covering all aspects of the communications field. Telecommunications technology is progressing at a rapid pace, and the IEEE Communications Society is committed to providing researchers and other professionals the information and tools to stay abreast. IEEE Communications Surveys and Tutorials focuses on integrating and adding understanding to the existing literature on communications, putting results in context. Whether searching for in-depth information about a familiar area or an introduction into a new area, IEEE Communications Surveys & Tutorials aims to be the premier source of peer-reviewed, comprehensive tutorials and surveys, and pointers to further sources. IEEE Communications Surveys & Tutorials publishes only articles exclusively written for IEEE Communications Surveys & Tutorials and go through a rigorous review process before their publication in the quarterly issues.

A tutorial article in the *IEEE Communications Surveys & Tutorials* should be designed to help the reader to become familiar with and learn something specific about a chosen topic. In contrast, the term ~~survey, as applied here, is defined to mean a survey of the literature. A survey article in IEEE~~ Communications Surveys & Tutorials should provide a comprehensive review of developments in a selected area, covering its development from its inception to its current state and beyond, and illustrating its development through liberal citations from the literature. Both tutorials and surveys should be tutorial in nature and should be written in a style comprehensible to readers outside the specialty of the article.

survey, as applied here, is defined to mean a survey of the literature. A survey article in IEEE Communications Surveys & Tutorials should provide a comprehensive review of developments in a selected area, covering its development from its inception to its current state and beyond, and illustrating its development through liberal citations from the literature. Both tutorials and surveys

论文分类

■ 科技论文

- 科技论文是指报道自然科学技术和技术开发创新性工作成果的论述文章，是以科技新成果为对象，采用科技语言和科学逻辑思维方式，并按照一定的写作格式撰写，经过正规严格的审查后公开发表的论文。
- 是科研工作者研究成果的深入具体展现，也是思想交流的主要桥梁。

论文分类——科技论文

■ 科技论文写作目的

- 详细报告自己的研究成果
- 对于某一领域的研究现状发表自己的观点和看法，或者针对某一已发表论文成果提出质疑
- 与同领域研究者进行学术交流，研究成果接受来自同行的审议，在互相探讨和争论中不断提升

■ 科技论文主要分为以下两类

- **Research paper**（研究型论文）
- **Letter**（快报）和 **Communication**（通讯论文）

论文分类——科技论文

■ Research paper

- 研究型论文，是反映原始研究工作的详细报告，其结论代表着人们在理解某一重要问题上所取得的一个实质性进展。研究者通常把撰写研究型论文看作是研究过程的最后步骤：形成研究的最终成果。可以是课题研究的终结报告，也可以是针对某一理论问题进行探讨、分析的学术论文。
- 这类论文应具有新的观点，新的分析方法和数据或结论，并具有科学性。内容应提供新的科技信息，有所发现，有所发明，有所创新，而不是重复模仿、抄袭前人的工作
- 研究型论文的涵盖范围较广，按照研究方式和内容来区分，可分为以下几类
 - 实验研究和分析类
 - 理论分析与推导类
 - 技术设计与创新类

论文分类——科技论文

■ 实验研究和分析类

特点

- 以实验和数据来说明和解释问题。
- 追求可靠的理论依据，具体、清楚、科学客观的实验设计和研究过程，合理准确的数据收集和处理过程，以及科学严密的分析论证结论，能够合理地解释研究结果。

主要结构

- 引言（Introduction）：阐述研究背景或者点明研究对象，介绍论文的重点内容或研究范围
- 方法（Method）：介绍实验过程和实验方法
- 结论和讨论（Results and Discussion）：展示研究结果和介绍结论，说明研究的借鉴意义，价值或贡献

论文分类——科技论文

■ 实验研究和分析类

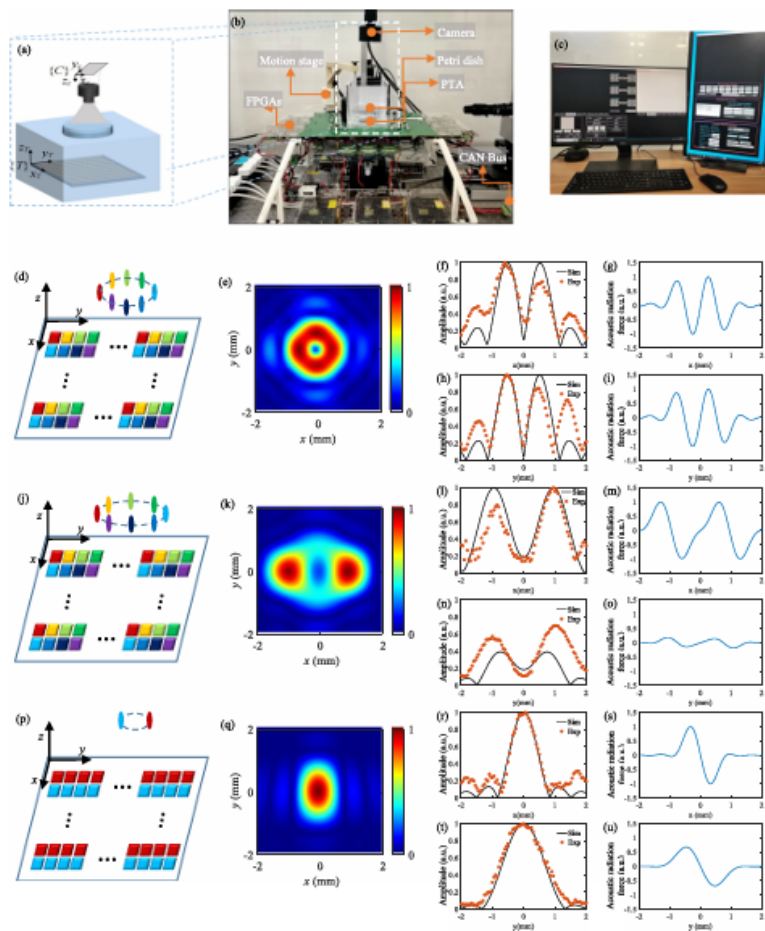


Fig. 1. (a) Diagram of the main component of the platform. (b) Optical image of experiment setup, which primarily consists of a phase transducer array, a microscope camera, and a host computer. (c) The control program and interactive interface of the host computer. (d), (e), and (f) Scheme of generating vortex trap field, asymmetric force field with low pressure in center, and asymmetric force field with high pressure in center. (g), (h), (i), (j), (k), (l), (m), (n), (o), (p), (q), (r), (s), (t), (u) Simulated corresponding field in x-y plane. (v), (w), (x), (y), (z), (aa), (ab), (ac), (ad), (ae), (af), (ag), (ah), (ai), (aj), (ak), (al), (am), (an), (ao), (ap), (aq), (ar), (as), (at), (au) Simulated and scanned result of beams profile. (v), (w), (x), (y), (z), (aa), (ab), (ac), (ad), (ae), (af), (ag), (ah), (ai), (aj), (ak), (al), (am), (an), (ao), (ap), (aq), (ar), (as), (at), (au) Simulated acoustic radiation force of corresponding field.

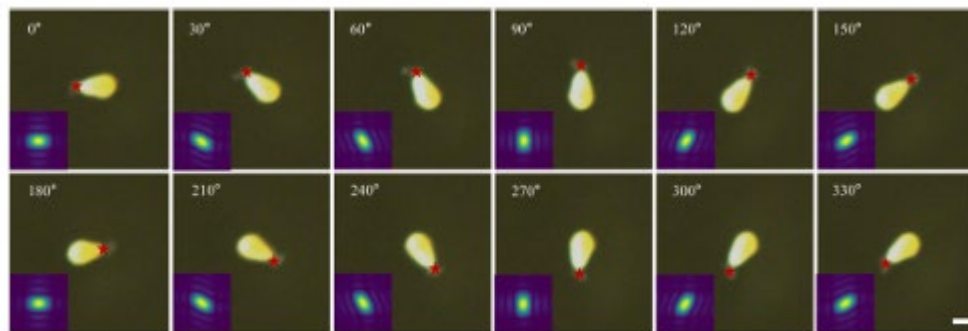


Fig. 6. Brine shrimp larvae orientation from 0° to 330° by interval of 30°. The orientation experiment is shown in Movie S5. Scale bar: 200 μm .

通过大量实验对比分析，作者提出的声学控制系统和声场控制方法能够对异质混合物中的生物体进行选择、稳定、精确、实时和培养皿内操作。

论文分类——科技论文

■ 理论分析与推导类

特点

- 通过数学推导或者逻辑推理得到新的理论。要求科学准确，逻辑严密的论证过程。
- 对于具体的新的系统、原理和模型等进行详尽的理论分析（比如系统性能分析），对过去的理论进行修正、补充和完善。要求分析论证和数学推导足够严谨正确，资料完善，且有实验或者仿真佐证。

主要结构

- 引言（Introduction）：阐述研究背景或者点明研究对象，介绍论文的重点内容或研究范围
- 方法（Method）：介绍理论提出和推导过程，以及所采用的数学工具
- 结论和讨论（Results and Discussion）：展示实验或仿真结果，证明其验证了提出的理论和分析的正确性

论文分类——科技论文

理论分析与推导类

POLYANSKIY *et al.*: FINITE BLOCKLENGTH REGIME

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where

$$B(\ell, w, n) = \sum_{w/2 \leq t \leq \min\{\ell, w\}} \binom{w}{t} \binom{n-w}{\ell-t}. \quad (10)$$

A $[k, n]$ linear code is generated by a $k \times n$ binary matrix. We can average (9) over an equiprobable ensemble of such matrices. Applying Jensen's inequality to pass expectation inside the minimum and noticing that $\mathbb{E}[A_w] = 2^{k-n} \binom{n}{w}$ we obtain the following achievability bound.

Theorem 5: For a BSC with crossover probability δ there exists a $[k, n]$ linear code such that a maximum likelihood decoder has a maximal probability of error P_e satisfying

$$P_e \leq \sum_{\ell=0}^n \delta^\ell (1-\delta)^{n-\ell} \times \min \left\{ \binom{n}{\ell}, \sum_{w=0}^n 2^{k-n} \binom{n}{w} B(\ell, w, n) \right\} \quad (11)$$

where $B(\ell, w, n)$ is given by (9).

A negligible improvement to (11) is possible if we average (9) over an ensemble of all full-rank binary matrices instead. Another modification by expurgating low-weight codewords [18] leads to a tightening of (11) when the rate is much lower than capacity.

For the BEC the results of [19, Th. 9] can be used to compute the exact value of the probability of error over an ensemble of all linear codes generated by full-rank $k \times n$ binary matrices [20].

Theorem 6 (Ashikhmin): Given a BEC with erasure probability δ , the average probability of error over all binary $k \times n$ linear codes with full-rank generating matrices chosen equiprobably is equal to (12), shown at the bottom of the page, where

$$\left[\begin{matrix} a \\ r \end{matrix} \right] \triangleq \prod_{j=0}^{r-1} \frac{2^a - 2^j}{2^r - 2^j}$$

is the number of r -dimensional subspaces of $\mathbb{F}_2^a$.

C. Achievability Bounds With Codeword Constraints

Suppose that all codewords are required to belong to some set $F \subset \mathcal{A}$. For example, there might be a cost $c(x)$ associated with using a particular input vector x , in which case the set F might be chosen as

$$F = \{x : c(x) \leq P\}. \quad (13)$$

A cost-constrained generalization of (5) due to Thomasian [21] (see also [22]) in which all the codewords are constrained to belong to F is

$$M \geq \beta (\epsilon - \mathbb{P}[i(X; Y) \leq \log \beta] - P_X[F]). \quad (14)$$

A cost-constrained version of (6) is

$$\epsilon \leq \frac{M-1}{\beta} + \mathbb{P}[i(X; Y) \leq \log \beta] + P_X[F]. \quad (15)$$

It should be noted that in both (14), and (15), the auxiliary distribution P_X is not constrained to take values on F . Theorem 3 admits the following generalization to the setting with cost constraints.

Theorem 7 (Gallager, With Cost): Suppose P_X is such that

$$\mathbb{E}[c(X)] \leq P \quad (16)$$

and denote

$$\mu(\delta) \triangleq \mathbb{P}[P - \delta \leq c(X) \leq P]. \quad (17)$$

Then, for any $\delta \in [0, P]$ such that $\mu(\delta) > 0$, $\lambda \in [0, 1]$ and $r \geq 0$ there exists an (M, ϵ) -code (average probability of error) with codewords in F given by (13) and such that we have (18), shown at the bottom of the page, where $P_{XY\tilde{X}}(a, b, c) = P_X(a)P_{Y|X=a}(b)P_X(c)$.

D. Converse Results

The simplest upper bound on the size of a code as a function of the average error probability follows from Fano's inequality:

Theorem 8: Every (M, ϵ) -code (average probability of error) for a random transformation $P_{Y|X}$ satisfies

$$\log M \leq \frac{1}{1-\epsilon} \sup_X I(X; Y) + \frac{1}{1-\epsilon} h(\epsilon) \quad (19)$$

Then by Fubini's theorem, the function $\phi : \mathcal{B} \mapsto \mathbb{R}_+$ is also positive measurable. Moreover

$$0 \leq \phi(y) \leq \int_G 1 \mu(dg) = 1. \quad (380)$$

Define for convenience

$$Q_Y^x \triangleq P_{Y^n | X^n = x}. \quad (381)$$

Then

$$\begin{aligned} \int_{\mathcal{B}} \phi(y) Q_Y^x(dy) &= \int_{\mathcal{B}} Q_Y^x(dy) \int_G (f \circ H)(g, y) \mu(dg) \\ &= \int_G \mu(dg) \int_{\mathcal{B}} (f \circ H)(g, y) Q_Y^x(dy). \end{aligned} \quad (382) \quad (383)$$

Change of the order of integration is possible by Fubini's theorem because $f \circ H$ is a bounded function. By the change of variable formula

$$\begin{aligned} \int_G \mu(dg) \int_{\mathcal{B}} (f \circ g)(y) Q_Y^x(dy) \\ = \int_G \mu(dg) \int_{\mathcal{B}} f(Q_Y^x \circ g^{-1})(y) dy. \end{aligned} \quad (384)$$

By the definition of Q_Y^x we have, for every measurable set E , $Q_Y^x[E] = Q_Y^0[E - x]$ and the measure Q_Y^0 is fixed under all isometries of $\mathbb{R}^n$:

$$\forall g \in G : Q_Y^0[gF] = Q_Y^0[g(F)]. \quad (385)$$

But then

$$(Q_Y^x \circ g^{-1})[E] \triangleq Q_Y^0[g^{-1}(E)] \quad (386)$$

$$= Q_Y^0[g^{-1}(E) - x] \quad (387)$$

$$= Q_Y^0[g^{-1}[E - g(x)]] \quad (388)$$

$$= Q_Y^{g(x)}[E]. \quad (389)$$

This proves that

$$Q_Y^x \circ g^{-1} = Q_Y^{g(x)}. \quad (390)$$

It is important that $x \in F_n$ implies $g(x) \in F_n$. In general terms, without AWGN channel specifics, the above argument shows that in the space of all measures on $\mathcal{B}$ the subset $\{Q_Y^x, x \in F_n\}$ is invariant under the action of G .

But $f \in \mathcal{F}_r$ and thus $\int f dQ_Y^x \geq \tau$ for every $x \in F_n$. So, from (384) and (390), we conclude

$$\int_{\mathcal{B}} \phi dQ_Y^x \geq \int_G \tau \mu(dg) = \tau. \quad (391)$$

➤ **理论推导：**推导了给定块长度和错误概率下可以达到的最大信道编码速率。

$$P_e = \sum_{i=0}^n \binom{n}{i} \delta^{n-i} (1-\delta)^i \sum_{r=\max\{0, k-n+i\}}^{\min\{k, i\}} \left[\begin{matrix} i \\ r \end{matrix} \right] \left[\begin{matrix} n-i \\ k-r \end{matrix} \right] \left[\begin{matrix} n \\ k \end{matrix} \right]^{-1} 2^{r(n-i-k+r)} (1-2^{r-k}) \quad (12)$$

$$\epsilon \leq M^\lambda \left(\frac{\exp(r\delta)}{\mu(\delta)} \right)^{1+\lambda} \mathbb{E} \left[\left(\mathbb{E} \left[\exp \left\{ \frac{i(\tilde{X}; Y)}{1+\lambda} + (c(\tilde{X}) - P)r \right\} \middle| Y \right] \right)^{1+\lambda} \right] \quad (18)$$

论文分类——科技论文

■ 技术设计与创新类

特点

- 建立新的模型，设计新的方案技术并进行全面的论证和分析，突出理论和技术优化方案的创新性。
- 数学模型的建立和参数设置合理，所设计的方案或者系统的有效性得到充分分析和论证，并且有其应用价值。

主要结构

- 引言（Introduction）：阐述研究背景或者点明研究对象，介绍论文的重点内容或研究范围
- 方法（Method）：介绍系统模型，优化方案设计以及分析
- 结论和讨论（Results and Discussion）：展示实验或仿真结果，证明所设计的技术方案或者新的系统可实现性能或成效优于现有技术和系统，具有应用价值

论文分类—科技论文

■ 技术设计与创新类

WANG *et al.*: ITERATIVE NOMA DETECTION FOR MULTIPLE ACCESS

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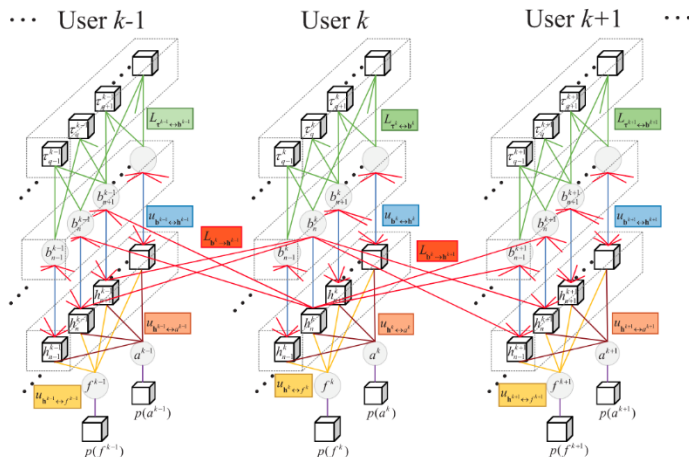


Fig. 3. The 3D factor graph model of asynchronous NOMA in satellite high-mobility communications. (1. Note that red lines indicate unidirectional message transmission and lines in other colors indicate dual-direction message transmission. 2. Note that b_n^k also has edges to channel function nodes h of users who are not adjacent to user k . Specially, b_n^k have edges to h_n^i and h_{n+1}^i for $i = 1, 2, \dots, k-1$, and have edges to h_n^i and h_{n-1}^i for $i = k+1, \dots, K$. This figure only draws part of these edges to keep neat.)

(except for the edge e) to the node α , summarized for the variable associated with e [33].

For the message passing schedule, passing messages are formulated from top to bottom of the factor graph. Let $L_{\tau_q^k \rightarrow b_n^k}^l$ denote the log-likelihood ratio (LLR) message passed from the LDPC check node (function node) of user- k τ_q^k to the chip node (variable node) of user- k b_n^k in the l iteration. According to the LDPC theory [34], $L_{\tau_q^k \rightarrow b_n^k}^l$ can be expressed as

$$L_{\tau_q^k \rightarrow b_n^k}^l = 2 \tanh^{-1} \left[\prod_{b_n^k \in N(\tau_q^k), v \neq n} L_{b_n^k \rightarrow \tau_q^k}^l \right], \quad (21)$$

where $N(\cdot)$ is the set of neighbors of a given node in the factor graph, and $b_n^k \in N(\tau_q^k)$, $v \neq n$ represents the neighbors of the LDPC check node τ_q^k except for the chip node b_n^k . Specially, $L_{b_n^k \rightarrow \tau_q^k}^l$ represents the LLR message sent from the chip node b_n^k to the LDPC check node τ_q^k in the iteration l , which can be expressed as

$$L_{b_n^k \rightarrow \tau_q^k}^l = L_{h_n^k \rightarrow b_n^k} + \sum_{\tau_v^k \in N(b_n^k), v \neq n} L_{\tau_v^k \rightarrow b_n^k}^l, \quad (22)$$

where $\tau_v^k \in N(b_n^k)$, $v \neq q$ represents the neighbors of the chip node b_n^k except for the LDPC check node τ_q^k , and $L_{h_n^k \rightarrow b_n^k}$ refers to the LLR message passed from the channel function node h_n^k to the chip node b_n^k in the iteration l , which is calculated in Eq. (41).

Based on the sum-product update rule, the LLR message sent from the chip node b_n^k to the channel function node h_n^k in the l iteration is the sum of LLR messages passed from the channel function node h_n^k and LLR messages passed from the LDPC check nodes τ_q^k that are neighbour to the chip node b_n^k , say,

$$L_{b_n^k \rightarrow h_n^k}^l = L_{h_n^k \rightarrow b_n^k} + \sum_{\tau_q^k \in N(b_n^k)} L_{\tau_q^k \rightarrow b_n^k}^l. \quad (23)$$

Since $L_{b_n^k \rightarrow h_n^k}^l$ satisfies

$$L_{b_n^k \rightarrow h_n^k}^l \triangleq \ln \left[\frac{u_{b_n^k \rightarrow h_n^k}^l(b_n^k = 1)}{u_{b_n^k \rightarrow h_n^k}^l(b_n^k = -1)} \right], \quad (24)$$

where $u_{b_n^k \rightarrow h_n^k}^l(b_n^k = 1)$ represents the reliability message of $b_n^k = 1$ passed from the chip node b_n^k to the channel function node h_n^k , and $u_{b_n^k \rightarrow h_n^k}^l(b_n^k = -1)$ represents the reliability message of $b_n^k = -1$ passed from the chip node b_n^k to the channel function node h_n^k in the iteration l . Thus, $u_{b_n^k \rightarrow h_n^k}^l(b_n^k = 1)$ and $u_{b_n^k \rightarrow h_n^k}^l(b_n^k = -1)$ can be further written as

$$u_{b_n^k \rightarrow h_n^k}^l(b_n^k = 1) = \frac{e^{L_{b_n^k \rightarrow h_n^k}^l}}{1 + e^{L_{b_n^k \rightarrow h_n^k}^l}}, \quad (25)$$

$$u_{b_n^k \rightarrow h_n^k}^l(b_n^k = -1) = \frac{1}{1 + e^{L_{b_n^k \rightarrow h_n^k}^l}}. \quad (26)$$

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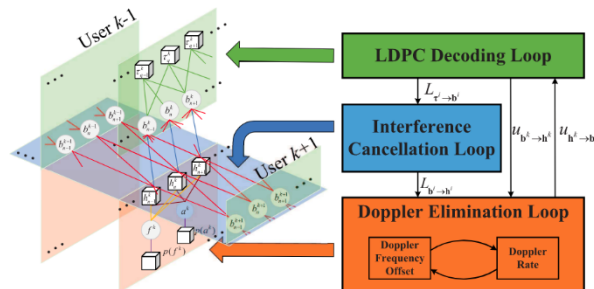


Fig. 4. The framework of turbo iterative asynchronous NOMA detection algorithm. (Note that b_n^k also has edges to channel function nodes h of users who are not adjacent to user k . Specially, b_n^k have edges to h_n^i and h_{n+1}^i for $i = 1, 2, \dots, k-1$, and have edges to h_n^i and h_{n-1}^i for $i = k+1, \dots, K$.)

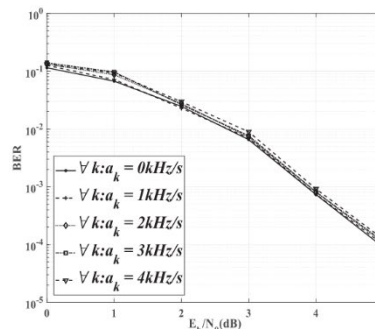


Fig. 6. BER performance of the proposed algorithm versus Doppler rates.

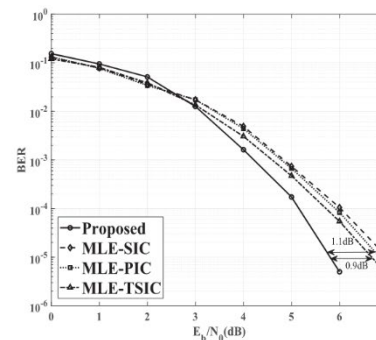


Fig. 8. BER performance versus different algorithms.

► **建立新模型：**作者构建了一种新的基于三维因子图的多用户检测方法。

论文分类——科技论文

■ Letter or Communication

快报和通讯论文非常相似，都必须具备**突出的创新点和较高的质量**。作者需要深入了解相关领域的研究前沿和热点，且要求明确该工作的学术意义以及可能产生的影响。通讯文章要求在有限的篇幅内阐述清楚一项完整的研究。

特点和优势：

- 简短 (Short): 篇幅短，一般少于3000字，研究成果能够在特定研究领域或更广泛的领域内产生一定的影响，更容易引起同行的关注或阅读兴趣
- 简要 (Brief): 只体现**最有趣或最突出的观点**，更多细节问题将在后续论文中深入讨论
- 快速 (Rapid): 发表周期快，在某些期刊中，同行评议过程可能直接被编辑审查取代，从而加速审稿

论文分类—科技论文

■ Letter or Communication

A Linear Time Algorithm for the Optimal Discrete IRS Beamforming

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Xin Li, Xin Chen, and Zhi-Quan Luo, *Fellow, IEEE*

Abstract—It remains an open problem to find the optimal configuration of phase shifts under the discrete constraint for intelligent reflecting surface (IRS) in polynomial time. The above problem is widely believed to be difficult because it is not linked to any known combinatorial problems that can be solved efficiently. The branch-and-bound algorithms and the approximation algorithms constitute the best results in this area. Nevertheless, this work shows that the global optimum can actually be reached in linear time on average in terms of the number of reflective elements (REs) of IRS. The main idea is to geometrically interpret the discrete beamforming problem as choosing the optimal point on the unit circle. Although the number of possible combinations of phase shifts grows exponentially with the number of REs, it turns out that there are only a linear number of circular arcs that possibly contain the optimal point. Furthermore, the proposed algorithm can be viewed as a novel approach to a special case of the discrete quadratic program (QP).

Index Terms—Discrete beamforming for IRS/IRS, global optimum, linear time algorithm, discrete quadratic program.

I. INTRODUCTION

CONFIGURING intelligent reflecting surface (IRS) concerns with the choice of a set of “best” phase shifts across reflective elements (REs), namely passive beamforming, to boost the signal reception at the target receiver. The existing studies in this area [1], [2] mainly divide into two categories: those assuming continuous phase shifts to ease optimization or analysis, and those restricting the phase shifts to a given discrete set from a practical standpoint. This work belongs to the latter. We establish a result that seems at first surprising: the discrete IRS beamforming problem has an $O(N)$ algorithm for its solution, where N is the number of REs.

The proposed algorithm stems from a geometric interpretation of the IRS beamforming problem which was first proposed in [3]. While [3] focuses on the binary beamforming case with each phase shift being either 0 or π , this work goes further to account for a general K -ary case where each phase shift takes

on K discrete values. We propose visualizing each possible solution as a point on the unit circle; the point moving around the unit circle captures the different beamforming decisions. The essential power of the above geometric method arises from the following observation: although the number of possible combinations of phase shifts grows exponentially with the number of REs, there are at most a linear number of unit circular arcs that possibly contain the point of the optimal beamforming. Furthermore, the utilities of all these arcs can be computed in linear time. The main idea behind the proposed algorithm is to try out all possible arcs and pick the best, and then recover the optimal phase shifts.

The existing works seldom consider global optimum for the discrete IRS beamforming. Although the IRS technology has been extensively studied over the past few years for many complicated system models, the simplest point-to-point model is still far less well understood. A common practice in the literature is to first solve the relaxed continuous beamforming problem via standard optimization, and then round the solution to the discrete set, e.g., [4], [5] based on semidefinite relaxation (SDR) and [6], [7] based on minorization-maximization (MM). For the relaxed continuous beamforming problem, [8] enforces the discrete constraint by penalization, but the resulting nonconvex problem is still difficult to deal with. Another popular idea [9], [10] is to decide the phase shift for one RE at a time, but the performance of such greedy search is not provable. Some other works [11], [12] resort to the branch-and-bound algorithm to obtain the global optimum of the discrete IRS beamforming. But these methods incur exponential time complexities and provide limited insights. For the practical approach, the best result so far in the literature is to attain an approximation ratio of the global optimum, e.g., [10] gives an approximation ratio of $\cos^2(\pi/K)$ in terms of the optimal signal-to-noise ratio (SNR), where K is the number of different values each phase shift can take on, and further [3] improves it to $0.5 + 0.5 \cos(\pi/K)$. Moreover, [13], [14] consider the discrete IRS beamforming in the absence of channel information. To the best of our knowledge, the present work gives the first polynomial time algorithm for the discrete IRS beamforming with provable global optimality.

Furthermore, it is worth mentioning the connection to the quadratic program (QP) like $\max x^T Q x$. The authors of [15] already observed that the discrete IRS beamforming problem at the link level boils down to solving a discrete QP—which is NP-hard in general [16] but can be efficiently solved under certain conditions [17], e.g., when each entry of x is selected from a binary set $\{0, 1\}$ and $\text{rank}(Q) \leq 2$. Actually, the

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- 篇幅简短
- 发表周期快

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论文分类——科技论文

■ Letter or Communication

I. INTRODUCTION

IN RECENT years, satellite communication (SATCOM) networks have been used, not only as the means of com-

... ..

II. SUPPOSED ENVIRONMENT AND PROBLEM STATEMENT

A. System Configuration

Fig. 1 shows an illustration of the main components of the HTS communication system. In this letter, the HTS is

... ..

III. PROPOSED POWER RESOURCE ALLOCATION WITH MULTI-BEAM DIRECTIVITY CONTROL

In this section, we construct a new mathematical model necessary for solving the trade-off problem and propose a

... ..

IV. PERFORMANCE EVALUATION

In this section, we quantitatively evaluate the relationship between the position of the traffic concentration area and

... ..

V. CONCLUSION

This letter constructed a new mathematical model necessary to solve the trade-off between transmit power and beam

- **结构简要：**具备清晰的结构，包含引言，问题阐述，问题建模，仿真，结论五部分。
- **准确性和创新性：**有限篇幅中通过建立模型并做出仿真对比实验，阐述研究的创新性。

论文分类

北京航空航天大学 研究生撰写学位论文的规定

(2009 年 7 月修订)

■ 学位论文

- 学位论文是表明作者从事科学研究取得创造性的成果或有了新的见解，并以此内容撰写而成，作为提出申请授予相应的学位是评审用的学术论文。
- 类似于科技论文，学位论文的结构同样包括选题，引言与背景介绍，正文，结论与参考文献。

学位论文是表明作者从事科学研究取得的创造性成果和创新见解，并以此为内容撰写的、作为申请学位时评审用的学术论文。

硕士学位论文应表明作者在本门学科上掌握了坚实的基础理论和系统的专门知识，对所研究的课题有新的见解，并具有从事科学研究工作或独立担负专门技术工作能力。

博士学位论文应表明作者在本门学科上掌握了坚实宽广的基础理论和系统深入的专门知识，在科学和专门技术上做出了创造性的成果，并具有独立从事科学研究工作的能力。

中华人民共和国国家标准

科学技术报告、学位论文和
学术论文的编写格式

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学位论文是表明作者从事科学研究取得创造性的结果或有了新的见解，并以此为内容撰写而成，作为提出申请授予相应的学位时评审用的学术论文。

论文分类——学位论文

■ 硕士学位论文

- 硕士研究生申请硕士学位要提交的论文。在导师指导下完成，必须具有一定程度的创新性，且由作者独立思考完成。
- 硕士学位论文只要求在某方面有改进和新见解。有一定的工作量，篇幅不能太短。

论文分类——学位论文

■ 博士学位论文

- 申请博士学位时提交的论文，可以是一篇论文的成果，但多数情况下是几篇论文成果的综合。
- 博士论文应能够反映作者坚实广博的科研理论基础知识 and 专业知识，具备独立从事科学技术研究工作的能力，反映出该科学技术领域最前沿的独创性成果。

论文分类——学位论文

□ 硕士学位论文与博士学位论文的区别

- 很多时候，硕士学位论文篇幅较博士学位论文小（但也要有一定篇幅），选题也会比较精细，一般只在比较小的方向上进行研究。
- 与硕士学位论文相比，博士学位论文是一本更加系统且独立的**著作**。包含课题研究历史现状，基础知识背景，系统模型或者实验设计，理论分析计算，实验、仿真或者实例讨论，未来与前景，参考文献，图表，附录等。

下面让我们看一下两篇学位论文的目录，感受硕士和博士学位论文的区别。

论文分类——学位论文

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一篇题为“压缩感知新模型及其在物联网中的应用研究”的硕士学位论文

论文分类——学位论文

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XI

一篇题为“异构蜂窝网络资源优化研究”的博士学位论文

论文分类——学位论文

■ 科技论文和学位论文的比较

- 科技论文主要面向于同领域研究者，结构较为简明精炼，只介绍主要的理论成果，直切主题。
- 与科技论文相比，学位论文为了体现作者的研究水平和能力，以供答辩评审，论文每部分的内容更加丰富和详尽，需要给出具体的实验过程或者理论推导过程，强调系统性，因此要求更清晰的论文逻辑结构。

课堂小结:

- 期刊分区的两种方式 and 它们的区别，以及查询分区的途径。
- 论文分类——综述论文。参考期刊：IEEE Communications Surveys & Tutorials; IEEE Communication Magazine。
- 论文分类——科技论文。Research paper: 实验研究和分析类；理论分析与推导类；技术设计与创新类；Letter or Communication。
- 论文分类——学位论文。硕士学位论文与博士学位论文。